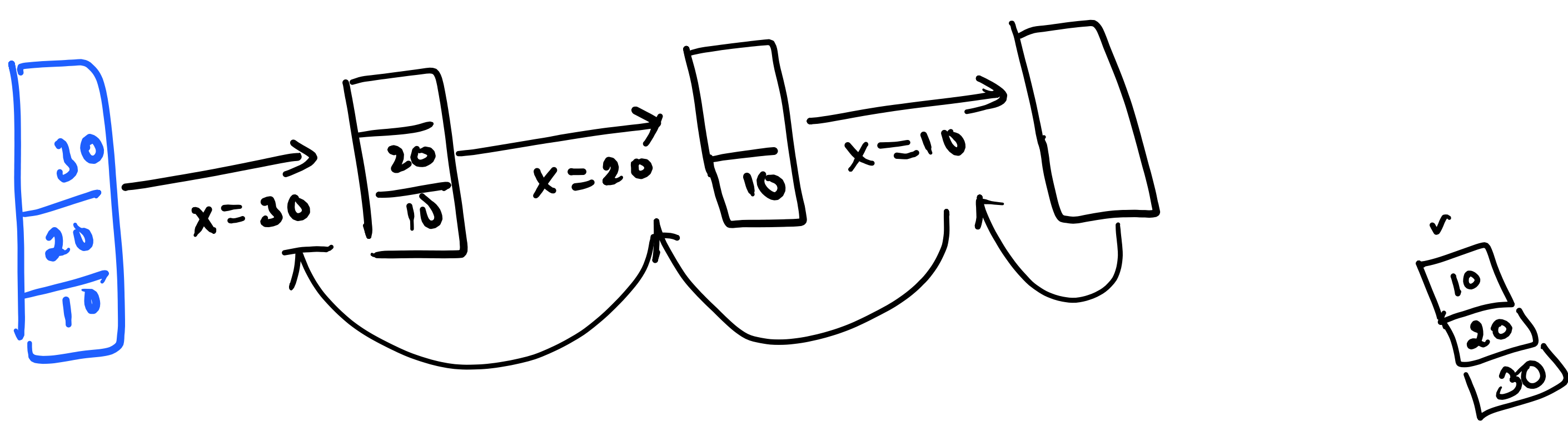
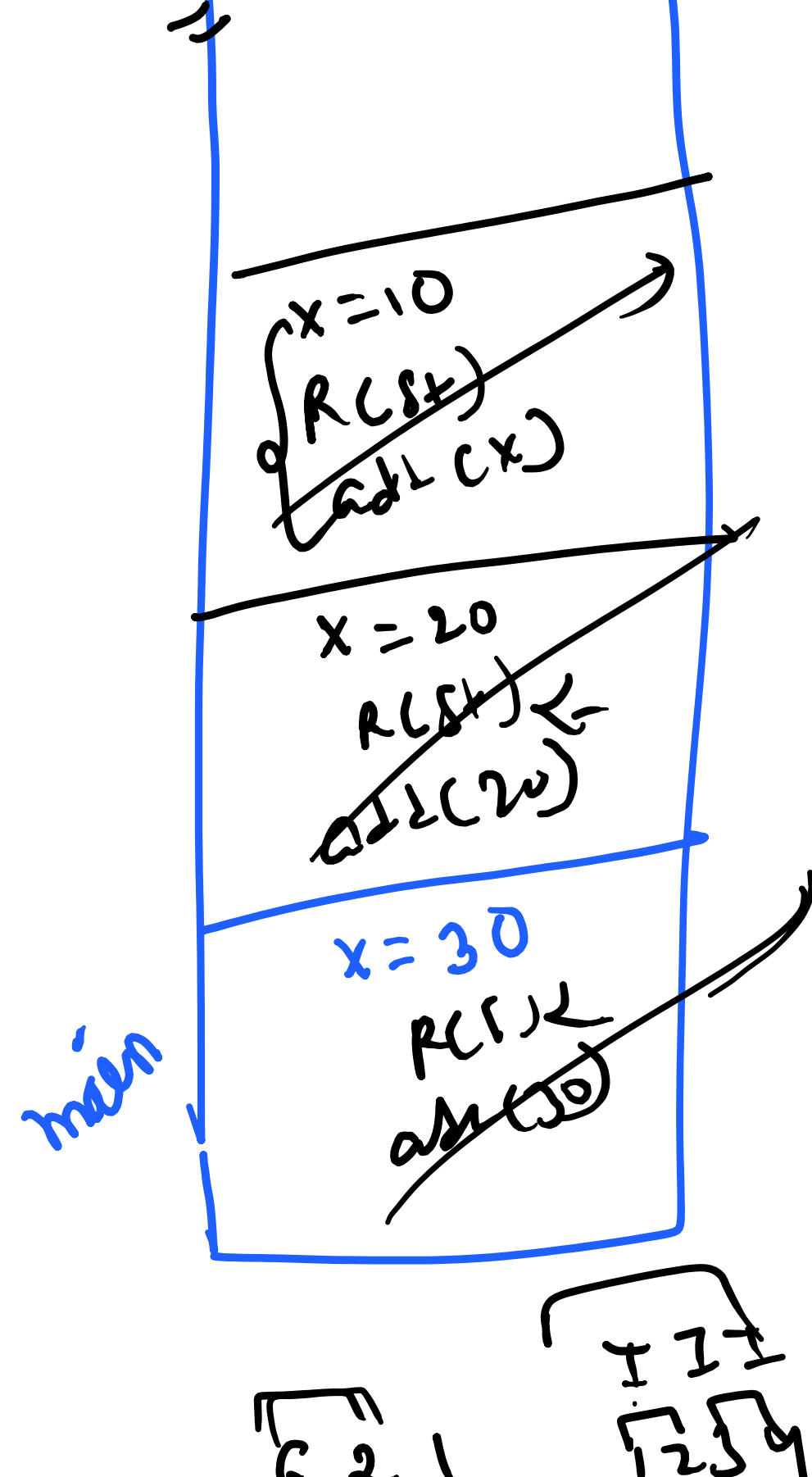




```
private static void Reverse(Stack<Integer> st) {
    // TODO Auto-generated method stub
    if (st.isEmpty()) {
        return;
    }
    int x = st.pop();
    Reverse(st);
    addlast(st, x);
}

public static void addlast(Stack<Integer> st, int item) {
    if (st.isEmpty()) {
        st.push(item);
        return;
    }
    int x = st.pop();
    addlast(st, item);
    st.push(x);
}
```



DDD
94321

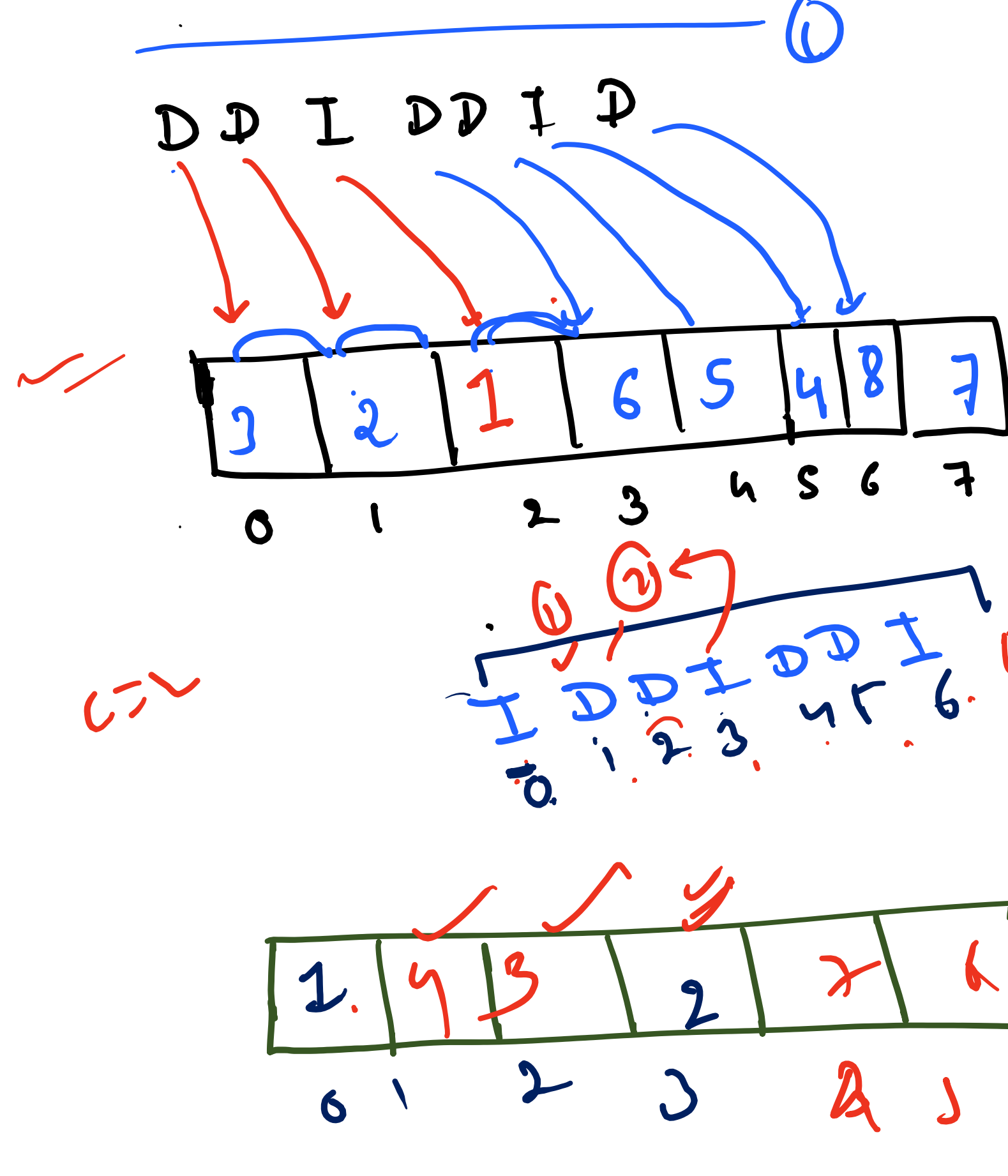
DD
321

DD
42

DD
62

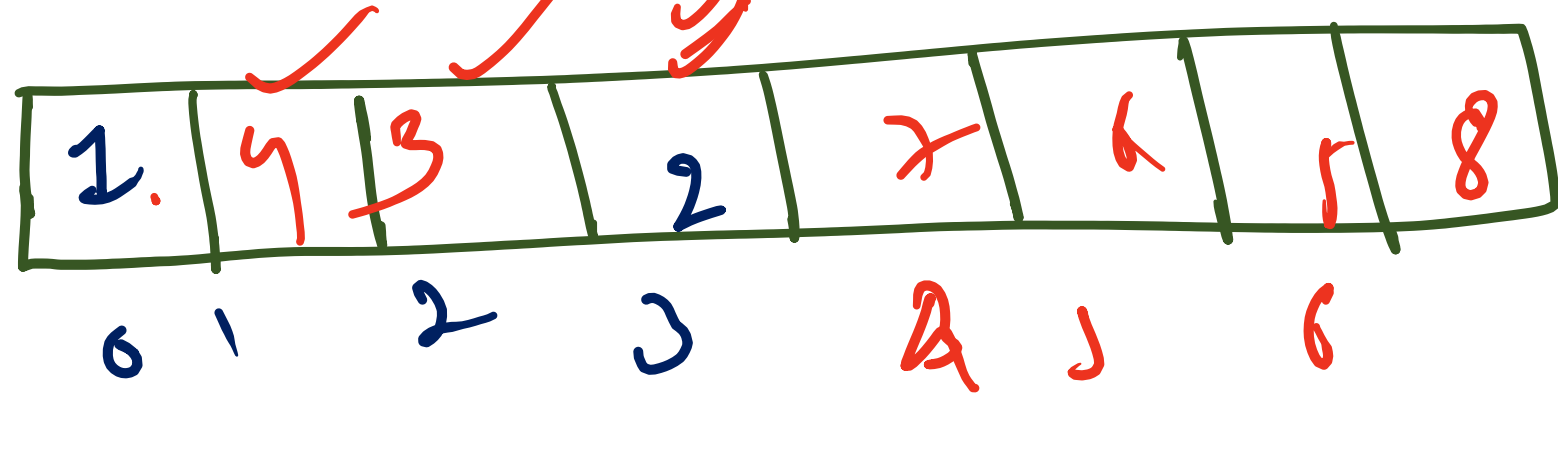
DD
734

108



i = s.length

112



7

7

Next Greater Element

Given an array of integers, for each element in the array, find the next greater element (NGE) in the array. The next greater element for an element x is the first element y to the right of x in the array such that y > x. If no such element exists, the NGE for that element is -1.

Input format

The first line contains an integer n, the size of the array.
The second line contains n space-separated integers A[0], A[1], ..., A[n-1].

Output format

Output a single line containing n space-separated integers, where each integer is the next greater element for the corresponding element in the input array. If no greater element exists, output -1.

203 1 7 8 11
-1 7 7 11 8 -1

0 1 2 3 4 5 6 7 8
[50, 5, 3, 2, 8, 7, 9, 11, 2]

next gen

for (ci = 0; ci < n; ci++)

while (!st.isEmpty() & arr[ci] < arr[st.peek()])

arr[st.pop()] = arr[ci]

3

= st.push(ci)

}

Stock Span

The stock span problem is a financial problem where we have a series of N daily price quotes for a stock and we need to calculate span of stock's price for all N days. You are given an array of length N, where ith element of array denotes the price of a stock on ith day. Find the span of stock's price on ith day, for every 1 ≤ i ≤ N.

A span of a stock's price on a given day, i, is the maximum number of consecutive days before the (i+1)th day, for which stock's price on these days is less than or equal to that on the ith day.

Input format

First line contains integer N denoting size of the array.
Next line contains N space separated integers denoting the elements of the array.

Output format

Display the array containing stock span values.

Example 1

Input
5
30
35
40
38
35
Output
1 2 3 1 1 END

0 1 2 3 4 5 6 7 8
[50, 5, 3, 2, 8, 7, 9, 11, 2]

0 1 2 3 4 5 6 7 8
[50, 5, 3, 2, 8, 7, 9, 11, 2]

for (ci = 0; ci < n; ci++)

while (!st.isEmpty() & arr[ci] < arr[st.peek()])

st.pop()

if (st.isEmpty())

arr[ci] = ci + 1

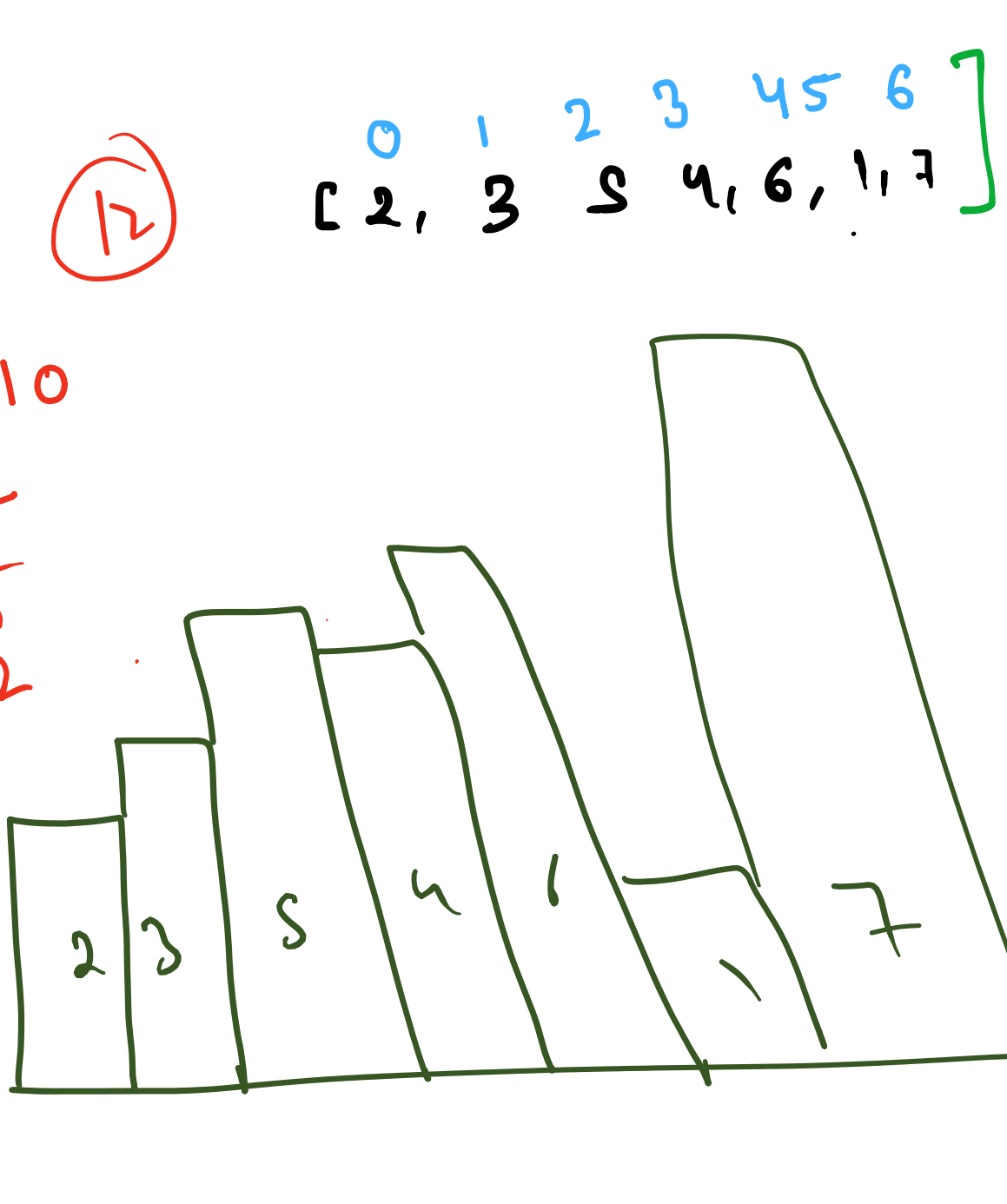
else

arr[ci] = i - st.peek()

st.push(ci)

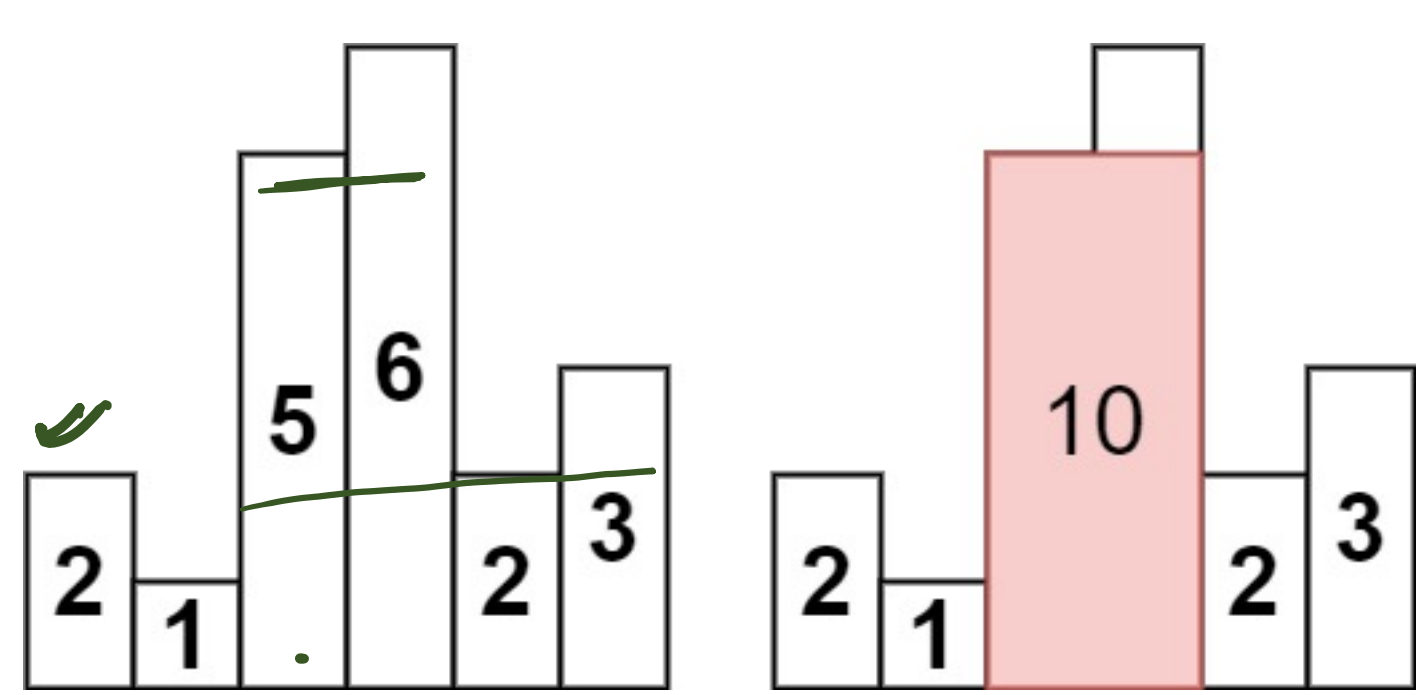
}

50
11
7
3



2x1=2
1x6=6
5x2=10
8x1=8
6x4=24
2x7=14

10



0 1 2 3 4 5 6
[2, 3, 5, 4, 6, 1, 7]

5-1-1=3

(R-L-1) x h =

R x h

R=5

R=3 L=1

(3-1-1) x 5 = 5

CS-3-1 x 6 = 6

CS-1-1 x 4 = 12

CS-0-1 x 3 = 12

CS x 2 = 10

5x1=5 8x2=16

0 1 2 3 4 5 6 7
[2, 3, 5, 4, 6, 1, 7, 8]

R=5

h=6 -> 6x(5-1)=24

h=4 -> 4x(5-1)=16

h=3 -> 3x(5-1)=12

h=2 -> 2x5=10

R=arr.length
while (!st.isEmpty())
h = arr[st.peek()]

for (ci = 0; ci < n; ci++)

while (!st.isEmpty() & arr[ci] < arr[st.peek()])

st.pop()

if (st.isEmpty())

area = ci + 1

else

area = ci - st.peek()

st.push(ci)

area = max(area, h x (ci - st.peek() - 1))

st.push(ci)

}

189. Rotate Array

Given an integer array nums, rotate the array to the right by k steps, where k is non-negative.

Example 1:
Input: nums = [1,2,3,4,5,6,7], k = 3
Output: [5,6,7,1,2,3,4]
Explanation:
rotate 1 steps to the right: [7,1,2,3,4,5,6]
rotate 2 steps to the right: [6,7,1,2,3,4,5]
rotate 3 steps to the right: [5,6,7,1,2,3,4]

Example 2:
Input: nums = [-1,-100,3,99], k = 2
Output: [3,99,-1,-100]
Explanation:
rotate 1 steps to the right: [99,-1,-100,3]
rotate 2 steps to the right: [3,99,-1,-100]